

Modeling the Impact of Industrial Brick Kilns on Lichen Biodiversity: A Poisson Regression Analysis

Project Type: Environmental Data Science | **Statistical Method:** Poisson GLM

Project Overview

This study investigates the ecological impact of industrial brick kilns on local biodiversity. I utilized **Lichens** as bioindicators—species that are highly sensitive to air quality—to quantify how industrial proximity and atmospheric Sulfur Dioxide levels drive localized extinction events.

Note: This project utilizes a simulated dataset to demonstrate the efficacy of Generalized Linear Models (GLM) in ecological modeling.

Methodology

The analysis was conducted on a dataset of 50 distinct observation quadrats. Since the outcome variable (Lichen Extinction Count) consists of non-negative integers, I employed a **Poisson Regression Model**. This approach is statistically more robust than traditional linear regression for count-based ecological data, ensuring more accurate predictions.

Key Statistical Findings

Predictor Variable	Coefficient (β)	IRR ($\exp(\beta)$)	P-value	Significance
Kiln Proximity (km)	-0.3810	0.6832	0.0001	Highly Significant
\$SO_2\$ Level (ppm)	-0.6319	0.5316	0.7166	Not Significant

- The 32% Risk Reduction:** The analysis revealed that Kiln Proximity is a primary driver of lichen loss ($p < 0.001$). The Incident Rate Ratio (IRR) of **0.6832** demonstrates that for every 1 km increase in distance from an industrial kiln, the risk of localized lichen extinction decreases by approximately **32%**.
- Multicollinearity Note:** While \$SO_2\$ is a known pollutant, it did not show independent significance in this model, likely due to strong multicollinearity with kiln proximity.

Conclusion & Policy Implications

The study provides strong evidence that brick kilns significantly threaten local biodiversity. Based on these findings, I recommend the implementation of a **3–5-kilometer eco-buffer zone** between industrial brick kilns and sensitive ecosystems. This data-driven approach highlights the potential for predictive modeling to inform environmental zoning and sustainable land-use planning.